

Gas-Phase and Heterogeneous Equilibria

K_p, K_c and equilibria with solids and liquids

Name _____ Class _____ Date _____

EQUATIONS AND DATA

$$K_p = K_c(RT)^{\Delta n} \cdot \Delta n = \text{gas moles (products - reactants)} \cdot R = 0.08206 \text{ L}\cdot\text{atm}/(\text{mol}\cdot\text{K}) \cdot T(\text{K}) = t(^{\circ}\text{C}) + 273.15$$

Leave pure solids and liquids out of K · Q < K: forward · Q > K: reverse

Pressures in atm; K written without units

Count only gas moles for Δn. Show units for R and T.

A. Writing K

1. Write K_p for: (a) $2 \text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{SO}_3(\text{g})$ (b) $2 \text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$ (c) $\text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{O}(\text{g})$ (d) $\text{NH}_4\text{Cl}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$

2. Give Δn for each reaction in question 1, and say whether K_p is larger than, smaller than or equal to K_c at 25 °C.

B. Calculating K_p and K_c

3. At equilibrium, P(SO₂) = 0.20 atm, P(O₂) = 0.10 atm and P(SO₃) = 1.2 atm. Calculate K_p for reaction 1(a).

4. For reaction 1(a), $K_c = 280$ at 1000 K. Calculate K_p .

5. For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $K_p = 0.15$ at 25 °C. Calculate K_c .

C. Q and ICE tables

6. For $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, $K_p = 50$ at a certain temperature. A mixture contains H_2 and I_2 at 0.10 atm each and HI at 0.50 atm. Calculate Q and predict the direction of change.

7. Pure N_2O_4 at 1.00 atm is allowed to reach equilibrium at 25 °C ($K_p = 0.15$). Calculate the equilibrium partial pressures of N_2O_4 and NO_2 .

D. Challenge

8. Solid CaCO_3 is heated in a sealed container until $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ reaches equilibrium. Explain what happens to the pressure of CO_2 if (a) more CaCO_3 is added (b) some CO_2 is pumped out (at constant temperature).